

Sustainable Futures: A Gamified Virtual Reality Journey through the SDGs

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Abstract

‘Sustainable Futures: A journey through the SDGs’ is an educational Virtual Reality (VR) game designed to equip learners with the knowledge and competencies needed to address ongoing challenges related to the Sustainable Development Goals (SDGs). Serious games can engage students, develop their higher-order thinking, and foster retention and transfer of actionable skills (Hallinger et al., 2020). By navigating the virtual world and engaging in playful activities, learners are encouraged to rethink about their own actions and behaviours, and ultimately, improve their attitudes across the key dimensions of Sustainable Development (SD), namely Environmental, Social, and Economic. The game features visually appealing 3D graphics and utilises the playful nature of VR technology to immerse students into a unique learning experience. Upon launching the game, players enter an initial, orientation level to familiarize with the system controls and game mechanics, followed by three thematic levels dedicated to Environmental, Social, and Economic sustainability. Each level includes puzzles and challenges ranging from tree planting and recycling, to managing waste, designing a sustainable city using renewable energy, discovering secret codes, and unlocking levels. The final level aims to assess students’ knowledge and energise them to take action towards a more sustainable future. Initial testing and evaluation with Higher Education students provided useful insights on the game’s usability and students’ degree of enjoyment (Nisiotis et al., 2024). The findings can inform further research and development highlighting the role of educational technology in fostering a sustainable mindset (Leal Filho et al., 2025) and driving positive change.

Keywords: Education for Sustainable Development (ESD) · Sustainability mindset · Game-based learning · Virtual Reality (VR) · Higher Education

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